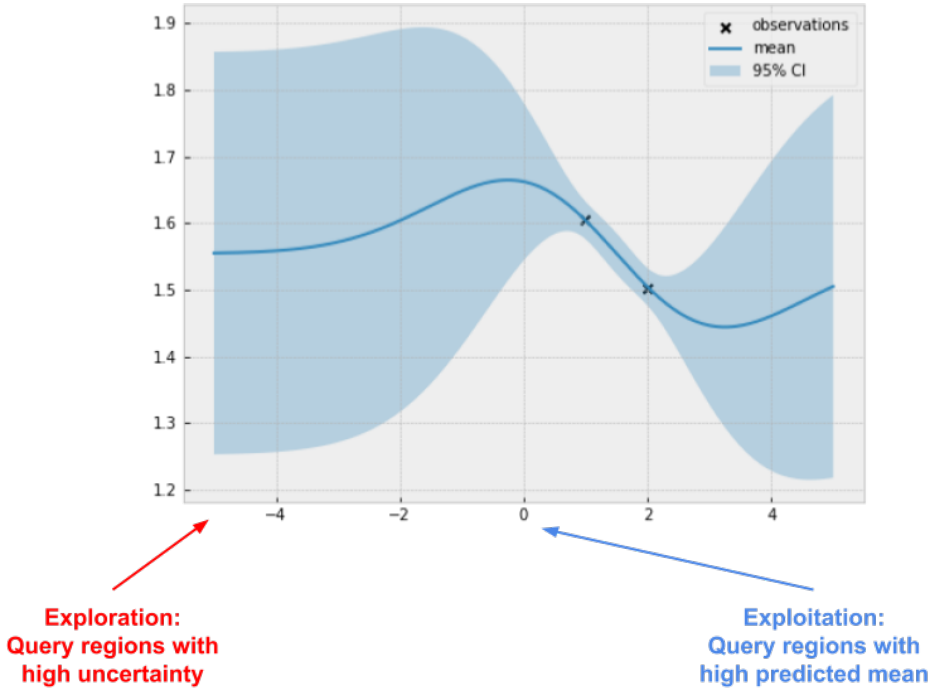


Figure 4.1 The exploration–exploitation tradeoff in Bayesian optimization. Each policy needs to decide whether to query regions with high uncertainty (exploration) or to query regions with high predicted mean

The exploration–exploitation tradeoff will set the stage for the different BayesOpt policies that we will be discussing in this and following chapters.

4.1.1 The Bayesian optimization loop and policies

First, let's remind ourselves what the BayesOpt loop looks like and the role BayesOpt policies play in this process. We will also implement a scaffold of this loop, which we will use to inspect BayesOpt policies in the future. Recall Figure 4.2, which is a repeat of figure 1.6 and shows how BayesOpt works on a high-level.

Specifically, BayesOpt is done via a loop that alternates between:

- Training a GP on the current training set. We have thoroughly covered how to do this in previous chapters.
- Using the trained GP to score data points in the search space in terms of how valuable they are in helping us identify the optimum of the objective function. The point that maximizes this score is selected to be labeled and added to the training data. How this scoring is done is determined by the BayesOpt policy used, and we will learn about different policies in this and Chapters 5 and 6.

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